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Lower Bound on the Overlattice-Based Sieve Algorithm

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Abstract

Lattice-based cryptography stands as one of the most pivotal candidates in post-quantum cryptography. To configure the parameters of lattice-based cryptographic schemes, a thorough comprehension of their concrete security is indispensable. Lattice sieving algorithms represent among the most critical tools for conducting concrete security analysis. Currently, the state-of-the-art BDGL-sieve (SODA 2016) achieves a time complexity of $2^{0.292n+o(n)}$, and Kirshanova and Laarhoven (CRYPTO 2021) have proven that the BDGL-sieve attains the lower bound under the technical paradigm of the Nearest Neighbor Search (NNS) problem. A natural question emerges: whether overlattice-based sieving algorithms (ANTS 2014) can outperform the BDGL-sieve within an alternative technical framework. This work provides an almost negative response to this question. Specifically, we propose a generalized overlattice tower model, which facilitates the proof of the lower bound for the overlattice-based method. Our findings indicate that the original Overlattice-sieve has already reached this lower bound. Consequently, the BDGL-sieve will maintain its status as the sieving algorithm with optimal time complexity, unless a revolutionary technical optimization is developed in the future.

Keywords: lattice cryptanalysis; sieving algorithms; lower bound; overlattice-based

1. Introduction

Over the past decade, driven by the unprecedented advancement of quantum computing, several problems that are computationally intractable under classical computing models have been demonstrated to allow for efficient solutions within the quantum computing paradigm. Most notably, Shor's algorithm [1] can factor large integers and solve the discrete logarithm problem in polynomial time, while Grover's algorithm [2] offers a quadratic speedup for unstructured search tasks. Such breakthroughs highlight the critical risk that traditional public-key cryptosystems may confront an existential crisis once large-scale practical quantum computers are realized, as these systems rely precisely on the hardness of the aforementioned problems for security.

Post-Quantum Cryptography. Against this backdrop, post-quantum cryptography (PQC) has emerged as a research priority, with diverse candidates including hash-based, code-based, multivariate polynomial-based, isogeny-based, and lattice-based cryptography. Among these, lattice-based cryptography has stood out as one of the most promising paradigms, particularly evidenced by its prominence in NIST's PQC standardization process [3]. Beyond its conjectured quantum attack resistance, lattice-based cryptography boasts distinctive merits that set it apart from other PQC candidates. It provides robust security assurances derived from the worst-case hardness of fundamental lattice problems [4,5],



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which is a property rarely found in other cryptosystems, and enables the construction of powerful and versatile cryptographic primitives [6–9].

Concrete Security of Lattice-Based Cryptosystems. To facilitate the practical implementation of lattice-based cryptographic schemes in real-world scenarios, researchers have placed growing emphasis on their concrete security in recent years. Distinct from asymptotic security, concrete security quantifies the actual computational effort required to break a scheme. The assessment of concrete security is predominantly dependent on the hardness of fundamental lattice problems [10–12], the most core of which are the Shortest Vector Problem (SVP), i.e., finding a non-zero lattice vector with the smallest Euclidean norm, and the Closest Vector Problem (CVP), i.e., finding the lattice vector closest to a given target vector in Euclidean space, along with their various variants. Consequently, understanding the computational complexity of these fundamental lattice problems is of paramount importance for the sound selection of security parameters in cryptographic primitives, ensuring they can withstand both classical and future quantum attacks.

1.1. Lattice Algorithms

Two major categories of approaches exist for tackling these lattice problems: approximate methods and exact methods.

Lattice Basis Reduction. The first category, approximate methods, aims to find vectors that are close to the optimal solution rather than being the exact one. Classic lattice basis reduction algorithms such as LLL [13] and its improved variant BKZ [14] are representative of this category. The LLL algorithm, proposed in 1982, revolutionized lattice research by efficiently outputting a reduced basis with guaranteed quality, while BKZ enhances its performance by incorporating blockwise reduction strategies. These algorithms are widely used in cryptographic analysis, but they often rely on the invocation of exact lattice problem solvers internally for block processing. This dependency naturally leads to the second category: exact algorithms, which seek to find the exact solution to lattice problems.

Currently, the mainstream exact algorithms for solving exact lattice problems primarily include enumeration methods and sieving methods.

Enumeration Methods. Lattice enumeration methods [15], dating back to the 1980s, work by systematically enumerating all the lattice points within a specific geometric region, typically a hyperball or intersection of cylinders, where a shortest non-zero lattice vector will be identified. While such methods exhibit favorable polynomial space complexity, making them feasible in resource-constrained environments, the state-of-the-art enumeration algorithms [16–18] still incur a super-exponential time cost of $2^{O(n \log n)}$, which becomes computationally prohibitive as the lattice dimension n increases.

Sieving Methods. Another class of exact algorithms, lattice sieving algorithms, addresses the time inefficiency of enumeration by trading off increased space usage for single-exponential time and space complexities, making them the preferred choice for solving high-dimensional exact lattice problems.

1.2. Frameworks of Lattice Sieving Algorithms

Currently, the two predominant sieving paradigms are the AKS-like and MV-like sieving frameworks, each having distinct design philosophies.

AKS-Like Sieving. The AKS-like sieving framework, introduced by Ajtai, Kumar, and Sivakumar [19], is typically structured around three core steps. First, it samples a large set of lattice points whose size scales as a single-exponential function of the lattice dimension. Subsequently, multiple rounds of pairwise reduction are performed on these lattice points according to specific distance-based rules. That is, if two vectors are sufficiently close, their difference will be shorter than either, and this shorter vector is retained while the longer

ones are discarded. Finally, after a polynomial number of iterative rounds, the algorithm is guaranteed to extract a shortest non-zero lattice vector with overwhelming probability.

MV-Like Sieving. In contrast, the MV-like sieving framework proposed by Micciancio and Voulgaris [20] adopts an alternative paradigm. It commences with the initialization of an empty vector list, followed by the iterative sampling of new lattice points. Each new lattice point undergoes reduction using the vector pool already stored in the list. In particular, it is subtracted by appropriate multiples of existing vectors to minimize its length, and the reduced vector is then incorporated into the list. This iterative process of sampling, reducing, and storing continues until a shortest vector is successfully identified.

Subsequently, a series of works [21–26] have proposed refinements based on these two sieving frameworks, driving continuous improvements in complexity. Under certain widely accepted heuristic assumptions, for example the Gaussian Heuristic, the current state-of-the-art sieving algorithm is the BDGL-sieve proposed by Becker, Ducas, Gama, and Laarhoven [27], which attains remarkable time and space complexities of $2^{0.292n+o(n)}$ and $2^{0.208n+o(n)}$, respectively. Furthermore, within the quantum computing model, Laarhoven [28] has leveraged Grover’s algorithm [2], further reducing the time complexity to $2^{0.265n+o(n)}$.

Overlattice-Based Sieving. At ANTS 2014, Becker, Gama, and Joux [29] proposed an alternative overlattice-based sieving method, which we refer to as the Overlattice-sieve algorithm. When addressing the exact SVP, the most notable distinction between this algorithm and the aforementioned AKS-like and MV-like frameworks lies in its expansion of the vector search space. The lattice vectors involved in each sieving round do not all belong to the original lattice $\mathcal{L}$. Instead, they are considered within a carefully constructed overlattice tower defined as $\mathcal{L} = \mathcal{L}_0 \subset \cdots \subset \mathcal{L}_k \subset \mathbb{R}^n$. Roughly speaking, the workflow of the overlattice-based sieving method leverages the hierarchical structure of the overlattice tower. The initial lattice vector list is populated with short lattice vectors sampled from the bottom lattice $\mathcal{L}_k$. In each subsequent round, short lattice vectors from the upper-layer lattice $\mathcal{L}_{i-1}$ are derived by leveraging short vectors from the lower-layer lattice $\mathcal{L}_i$. This process of descending the lattice tower continues until short lattice vectors belonging to the original target lattice $\mathcal{L} = \mathcal{L}_0$ are obtained. The Overlattice-sieve algorithm exhibits time and space complexities of $2^{0.3774n}$ and $2^{0.2925n}$, respectively [29]. It is less efficient than the BDGL-sieve, but with a unique advantage: it is inherently an exact CVP solver, capable of tackling both CVP and SVP, whereas the two mainstream sieving frameworks mentioned above are essentially specialized exact SVP solvers.

A natural and critical question arises in the field: Is there still room for further improvements in sieving algorithms, and can we be confident that the current state-of-the-art sieving algorithms will remain unrivaled in the decades to come? This question gains urgency as PQC standards solidify, requiring long-term security guarantees. At CRYPTO 2021, Kirshanova and Laarhoven [30] provided a partial answer by proving that the Locality-Sensitive Filters (LSF) technique employed in the BDGL-sieve has achieved the optimal asymptotic performance for the Nearest Neighbor Search (NNS) problem in the lattice cryptographic setting, which is a core subroutine in both AKS-like and MV-like sieving frameworks. Consequently, researchers can largely rule out the possibility of further asymptotic improvements that solely target the NNS subroutine within these two mainstream frameworks.

Unfortunately, the same level of theoretical clarity does not exist for the overlattice-based sieving algorithm. Its unique overlattice tower structure deviates from the NNS design of traditional sieves, leaving its complexity lower bounds poorly understood. One may thus reasonably inquire: *Could the overlattice-based approach, with further optimizations, eventually outperform the BDGL-sieve in terms of asymptotic complexity?* This paper addresses

this open question and demonstrates an almost negative answer, providing valuable theoretical guidance for the future direction of lattice sieve research.

1.3. Related Work

The evolution of lattice sieving algorithms can be traced back to the early 2000s, with the AKS-sieve marking the first major breakthrough. In 2001, Ajtai, Kumar, and Sivakumar [19] introduced the first lattice sieve algorithm having rigorous theoretical guarantees, namely the AKS-sieve. Rigorous complexity analysis demonstrates that its time and space complexities are $2^{5.90n+o(n)}$ and $2^{2.95n+o(n)}$, respectively [31], which is a significant improvement over the enumeration methods of the time, though still far from practical for high dimensions.

A notable innovation of the AKS-sieve lies in the introduction of a perturbation technique, enabling rigorous theoretical proofs regarding both its correctness and its complexity. However, this perturbation technique is computationally expensive in practice. Building upon the AKS-sieve framework, Nguyen and Vidick [31] dispensed with the computationally inefficient perturbation technique and devised a heuristic alternative known as the NV-sieve. While relying on heuristic assumptions about vector distribution, the NV-sieve achieved a dramatic leap in efficiency, with time and space complexities of $2^{0.415n+o(n)}$ and $2^{2.075n+o(n)}$, respectively.

In 2010, Micciancio and Voulgaris [20] developed an alternative family of lattice sieve algorithms that departed from the AKS-like sieve paradigm, including the List-sieve and Gauss-sieve. The List-sieve, which retains the rigorous nature of the AKS-sieve, achieves time and space complexities of $2^{3.20n+o(n)}$ and $2^{1.33n+o(n)}$, respectively. By contrast, the Gauss-sieve is a heuristic algorithm that adopts a more greedy reduction strategy similar to the classical Gauss reduction, and it can attain the same time and space complexities as the NV-sieve, while being more memory-efficient in practice.

Subsequently, researchers focused primarily on advancing heuristic sieve algorithms, as their practical performance made them more relevant for cryptographic security analysis. A key direction of improvement was optimizing the iteration approach of sieving rounds. The Level-sieve proposed by Wang et al. [21] and Zhang et al. [32] advanced the time and space complexities to $2^{0.378n+o(n)}$ and $2^{0.283n+o(n)}$, respectively. Its core idea, originating from the NV-sieve, was to divide each sieve iteration round into multiple hierarchical levels based on vector length. The total number of vector pairs that need to be checked across these multiple levels is significantly reduced, thereby lowering the overall time complexity.

Another critical innovation was shifting from pairwise vector reduction to multi-vector reduction. Building on the List-sieve, Bai, Laarhoven, and Stehle [22] proposed the Tuple-sieve, with time and space complexities reaching $2^{0.481n+o(n)}$ and $2^{0.189n+o(n)}$, respectively. A distinguishing feature of the Tuple-sieve is that it reduces k -tuples of vectors, for $k > 2$, at a time during the vector reduction process, rather than merely pairs. Subsequently, the HKL-Tuple-sieve proposed by Herold et al. [23,33] refined the process of finding short vectors in the Tuple-sieve by optimizing the tuple selection strategy and incorporating more efficient reduction rules. This further advanced the time and space complexities to $2^{0.330n+o(n)}$ and $2^{0.208n+o(n)}$, respectively, bringing the algorithm closer to practicality for high-dimensional lattices.

A parallel direction of improvement focused on accelerating the Nearest Neighbor Search (NNS) subroutine, which is often the bottleneck in sieve iterations. In 2015, the Hash-sieve introduced by Laarhoven [24] leveraged the Locality-Sensitive Hashing (LSH) technique to accelerate the brute-force search component of each sieve iteration round, and it was implemented based on the Gauss-sieve. In 2016, Becker, Ducas, Gama, and Laarhoven [27] proposed the BDGL-sieve, which employs the Locality-Sensitive Filters

(LSF) technique, a technique similar to LSH, to further accelerate the brute-force search phase. Its time and space complexities are $2^{0.292n+o(n)}$ and $2^{0.208n+o(n)}$, respectively, solidifying its position as the current state-of-the-art heuristic sieve algorithm. We summarize the complexities of the aforementioned sieving algorithms in Table 1. This table is sorted by time complexity.

Table 1. Summary of complexities of lattice sieving algorithms.

Algorithm	Authors	Conference/Journal (Year)	Time Complexity	Space Complexity
AKS-sieve [19]	Ajtai, Kumar, Sivakumar	STOC (2001)	$2^{5.90n+o(n)}$	$2^{2.95n+o(n)}$
List-sieve [20]	Micciancio, Voulgaris	SODA (2010)	$2^{3.20n+o(n)}$	$2^{1.33n+o(n)}$
Tuple-sieve [22]	Bai, Laarhoven, Stehlé	LMS J. Comput. Math. (2016)	$2^{0.481n+o(n)}$	$2^{0.189n+o(n)}$
NV-sieve [31]	Nguyen, Vidick	J. Math. Crypt. (2008)	$2^{0.415n+o(n)}$	$2^{0.075n+o(n)}$
Gauss-sieve [20]	Micciancio, Voulgaris	SODA (2010)	$2^{0.415n+o(n)}$	$2^{0.075n+o(n)}$
Level-sieve [21,32]	Wang et al.; Zhang et al.	AsiaCCS (2011); SAC (2013)	$2^{0.378n+o(n)}$	$2^{0.283n+o(n)}$
Overlattice-sieve [29]	Becker, Gama, Joux	ANTS (2014)	$2^{0.3774n+o(n)}$	$2^{0.2925n+o(n)}$
Hash-sieve [24]	Laarhoven	CRYPTO (2015)	$2^{0.337n+o(n)}$	$2^{0.208n+o(n)}$
HKL-Tuple-sieve [23,33]	Herold, Kirshanova, Laarhoven	PKC (2017); PKC (2018)	$2^{0.330n+o(n)}$	$2^{0.208n+o(n)}$
BDGL-sieve [27]	Becker, Ducas, Gama, Laarhoven	SODA (2016)	$2^{0.292n+o(n)}$	$2^{0.208n+o(n)}$

Beyond asymptotic complexity improvements, researchers have also focused on enhancing the practical implementation effectiveness of sieve methods. Techniques such as the progressive sieve [34], dimension-for-free technique [26], and the G6K framework [35] have all significantly optimized the practical execution performance of sieve methods. Among these, G6K has become one of the primary tools for analyzing the concrete security of lattice-based cryptographic schemes, widely used in evaluating candidates for NIST's PQC standardization.

1.4. Contributions

The landscape of lattice sieve algorithms has reached a critical point. Kirshanova and Laarhoven [30] have proven that the BDGL-sieve, which represents the culmination of AKS-like and MV-like sieves, has achieved the complexity lower bound of $2^{0.292n+o(n)}$ for NNS-based frameworks. This result closes the door on improvements to existing paradigms, shifting the focus to alternative approaches like the overlattice-based method. Against this backdrop, a natural open question emerges: Can the overlattice-based paradigm, with its unique ability to solve both the SVP and the CVP, achieve a better asymptotic complexity than the BDGL-sieve? This paper addresses this question by establishing the first formal lower bounds for overlattice-based sieving algorithms, making three key contributions to the field.

First, we provide the first rigorous-complexity lower bound analysis for the overlattice-based sieving paradigm. Prior work on the Overlattice-sieve [29] only reported its asymptotic complexity without addressing whether further improvements are possible. We fill this gap by demonstrating that its time complexity of $2^{0.3774n}$ coincides exactly with the lower bound for overlattice-based methods. This means that no overlattice-based algorithm can achieve a better asymptotic time complexity than the original Overlattice-sieve, unless a revolutionary change is made to the paradigm's core framework.

Second, we propose a generalized overlattice tower model that addresses a critical limitation of the original Overlattice-sieve: its overly simplistic overlattice tower structure. Our generalized model allows for variable layer indices, and adaptable radii for each layer, enabling a comprehensive analysis of how different tower parameters impact complexity. Through rigorous analysis of this model, we derive the lower bound and show that the parameters of the original Overlattice-sieve are optimal for minimizing complexity. This

model serves as a foundational tool for future research on overlattice-based methods, providing a framework to evaluate new algorithmic variants.

Third, our results strengthen the confidence in the BDGL-sieve as the gold standard for concrete security analysis of lattice-based cryptosystems. By combining Kirshanova and Laarhoven's lower bound for NNS-based sieves [30] with our new lower bound for overlattice-based methods, we confirm that the BDGL-sieve's complexity of $2^{0.292n+o(n)}$ represents a benchmark for exact SVP solvers across all major paradigms. This validation is critical for cryptographic practitioners, as it ensures that security parameters selected using the BDGL-sieve as a benchmark will remain secure against future algorithmic advances.

1.5. Paper Organization

The remainder of this paper is structured as follows. In Section 2, we establish the necessary mathematical background, including formal definitions of lattices, overlattices, and Gaussian Heuristics. In Section 3, we provide a succinct review of the overlattice-based sieving algorithm put forward by Becker, Gama, and Joux [29]. To tackle the problem that the parameters of the overlattice tower in the Overlattice-sieve are overly simplistic, we propose a generalized overlattice tower model in Section 4. Subsequently, leveraging this model, we conduct a rigorous and detailed analysis of the time complexity lower bounds for the overlattice-based sieving algorithm in Section 5. Finally, we present the conclusions of this paper in Section 6.

2. Preliminaries

All vectors are denoted by bold lower-case letters and are to be understood as column vectors. The euclidean norm of a vector $\mathbf{v}$ is denoted by $\|\mathbf{v}\|$. Matrices are denoted by bold capital letters. A matrix $\mathbf{B}$ can be seen as a collection of (column) vectors $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$. When $\mathbf{B} \in \mathbb{R}^{m \times n}$ has full rank n , the lattice $\mathcal{L}$ generated by the basis $\mathbf{B}$ is denoted by $\mathcal{L}(\mathbf{B}) = \{\mathbf{B}\mathbf{x} | \mathbf{x} \in \mathbb{Z}^n\}$.

We denote by $\mathbf{B}^* = (\mathbf{b}_1^*, \dots, \mathbf{b}_n^*)$ the Gram–Schmidt orthogonalization of a lattice basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$. In particular, the lattice basis $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ can be written uniquely as a product $\mathbf{B}^T = \mathbf{M} \cdot \mathbf{D} \cdot \mathbf{O}$, where

$$\mathbf{M} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \mu_{2,1} & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{n,1} & \dots & \mu_{n,n-1} & 1 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} \|\mathbf{b}_1^*\| & 0 & \dots & 0 \\ 0 & \|\mathbf{b}_2^*\| & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \|\mathbf{b}_n^*\| \end{pmatrix}, \mathbf{O} = \begin{pmatrix} \frac{\mathbf{b}_1^{*T}}{\|\mathbf{b}_1^*\|} \\ \frac{\mathbf{b}_2^{*T}}{\|\mathbf{b}_2^*\|} \\ \vdots \\ \frac{\mathbf{b}_n^{*T}}{\|\mathbf{b}_n^*\|} \end{pmatrix},$$

and $\mathbf{O}$ is an $n \times m$ matrix with orthonormal row vectors. The parameters $\mu_{i,j}$ are called the Gram–Schmidt coefficients. In particular, we set $\mu_{i,i} = 1$. The volume of a lattice $\mathcal{L}(\mathbf{B})$ is $\text{vol}(\mathcal{L}(\mathbf{B})) = \prod_i \|\mathbf{b}_i^*\|$, which is an invariant of a lattice and does not depend on the choices of basis.

If $\mathcal{L}_1 \subset \mathcal{L}_2$, we say that $\mathcal{L}_1$ is a sublattice of $\mathcal{L}_2$, and $\mathcal{L}_2$ is accordingly referred to as an overlattice of $\mathcal{L}_1$. Additionally, when we also have $\mathcal{L}_1, \mathcal{L}_2 \subset \mathbb{R}^n$, the quotient group $\mathcal{L}_2/\mathcal{L}_1$ forms a finite abelian group, whose order is given by $\text{vol}(\mathcal{L}_1)/\text{vol}(\mathcal{L}_2) = [\mathcal{L}_2 : \mathcal{L}_1]$.

For example, in Figure 1, the black crosses represent the lattice points of the lattice $\mathcal{L}_1$, the vectors $\mathbf{c}_1, \mathbf{c}_2$ being the lattice basis for $\mathcal{L}_1$. The blue dots denote the lattice points of the lattice $\mathcal{L}_2$, and the vectors $\mathbf{b}_1, \mathbf{b}_2$ form a lattice basis for $\mathcal{L}_2$. Since $\mathcal{L}_1 \subset \mathcal{L}_2 \subset \mathbb{R}^2$, $\mathcal{L}_1$ is a sublattice of $\mathcal{L}_2$, whereas $\mathcal{L}_2$ is an overlattice of $\mathcal{L}_1$. Additionally, the order of the quotient group $\mathcal{L}_2/\mathcal{L}_1$ is given by $\text{vol}(\mathcal{L}_1)/\text{vol}(\mathcal{L}_2) = [\mathcal{L}_2 : \mathcal{L}_1] = 2$.

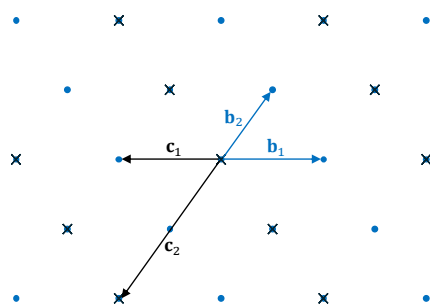


Figure 1. Sublattices and overlattices.

The first minimum of a lattice $\mathcal{L}$ is the length of a shortest non-zero vector, denoted by $\lambda_1(\mathcal{L})$. The well-known Short Vector Problem (SVP) requires finding a shortest non-zero vector $\mathbf{v}$ such that $\|\mathbf{v}\| = \lambda_1(\mathcal{L})$, given a lattice basis $\mathbf{B}$. On the other hand, another prominent problem is the Closest Vector Problem (CVP). Given a lattice basis $\mathbf{B}$ and a target point $\mathbf{t}$, the CVP entails finding a lattice point $\mathbf{v}$ that is closest to the target point $\mathbf{t}$, satisfying

$$\|\mathbf{v} - \mathbf{t}\| = \min_{\mathbf{w} \in \mathcal{L}} \|\mathbf{w} - \mathbf{t}\|.$$

Note that the target point $\mathbf{t}$ is generally not a lattice point.

The Gaussian Heuristic provides an estimate for the first minimum. In fact, the Gaussian Heuristic provides an estimate of the number of lattice points inside a “nice enough” measurable set.

Definition 1 (Gaussian Heuristic). *Given an n -dimensional lattice $\mathcal{L}$, if the volume of a set $S \subseteq \mathbb{R}^n$ is $\text{vol}(S)$, then the number of points in $S \cap \mathcal{L}$ is approximately $\text{vol}(S) / \text{vol}(\mathcal{L})$.*

Further, the length $\lambda_1(\mathcal{L})$ of the shortest non-zero vector of a lattice $\mathcal{L}$ is estimated to be the radius of the n -dimensional hyperball of volume $\text{vol}(\mathcal{L})$, i.e.,

$$\lambda_1(\mathcal{L}) \approx \text{gh}(\mathcal{L}) = \text{vol}(\mathcal{L})^{1/n} \cdot \text{vol}(\mathcal{B}_n(1))^{-1/n},$$

where $\text{vol}(\mathcal{B}_n(1))$ is the volume of the n -dimensional unit hyperball. In particular, the volume of an n -dimensional hyperball of radius r is

$$\text{vol}(\mathcal{B}_n(r)) = \frac{\sqrt{\pi}^n}{\Gamma(\frac{n}{2} + 1)} \cdot r^n,$$

where $\Gamma(\cdot)$ is the Gamma function. Therefore,

$$\text{vol}(\mathcal{B}_n(1))^{-1/n} = \sqrt{\frac{n}{2\pi e}} \cdot (1 + o(1)).$$

For high-dimensional random lattices, the estimates from the Gaussian Heuristic are generally quite accurate [14]. Moreover, given a lattice $\mathcal{L}$ and a nice set S , one can heuristically derive upper and lower bounds on the number of points in $\mathcal{L} \cap S$. That is, for some constant $G_H \geq 1$, we have

$$\frac{1}{G_H} \cdot \frac{\text{vol}(S)}{\text{vol}(\mathcal{L})} \leq |\mathcal{L} \cap S| \leq G_H \cdot \frac{\text{vol}(S)}{\text{vol}(\mathcal{L})}.$$

Through extensive experimental verification, Becker, Gama, and Joux [29] set the constant G_H to 2. Note that Gaussian Heuristics plays a pivotal role in determining both the complexity and correctness of the overlattice-based methods considered in Sections 3–5. Specifically,

it is widely employed for estimating the number of lattice points in the intersection of two balls in the Merge subroutine.

3. Revisiting the Overlattice-Based Sieve Algorithm

In this section, we briefly introduce the Overlattice-sieve algorithm proposed by Becker, Gama, and Joux [29], and meanwhile we identify the deficiency that the parameters of the overlattice tower in the Overlattice-sieve algorithm are overly simplified.

Ultimately, the Overlattice-sieve algorithm can yield a set C_0 of short vectors on an arbitrary coset $\mathbf{t} + \mathcal{L}$, encompassing almost all short vectors among the first β^n ones. This set C_0 can be exploited to solve a variety of lattice problems. For example, for solving the SVP, we set $\mathbf{t} = \mathbf{0}$, whereas for addressing the CVP, $\mathbf{t}$ is designated as $-\mathbf{t}'$, with $\mathbf{t}'$ representing the target vector specified by the CVP instance.

The Overlattice-sieve algorithm considers lattice points lying within a so-called overlattice tower

$$\mathcal{L} = \mathcal{L}_0 \subset \cdots \subset \mathcal{L}_k \subset \mathbb{R}^n,$$

where $k = O(n)$, and

$$\frac{\text{vol}(\mathcal{L}_i)}{\text{vol}(\mathcal{L}_{i-1})} = [\mathcal{L}_i : \mathcal{L}_{i-1}] = \lceil \alpha^n \rceil.$$

We denote the set of short vectors on each lattice coset as, for $i = 0, 1, \dots, k$,

$$C_i = (\mathbf{t}_i + \mathcal{L}_i) \cap \mathcal{B}(\mathbf{0}, R_i),$$

where the radius $R_i = \beta \cdot \text{gh}(\mathcal{L}_i)$ and $2\mathbf{t}_{i-1} = \mathbf{t}_i$, i.e., $\mathbf{t}_i := \frac{\mathbf{t}_0}{2^i}$. Under the Gaussian Heuristic, we have $|C_i| \approx \beta^n$.

The basic workflow of the Overlattice-sieve method is illustrated in Figure 2. In particular, the core steps are outlined as follows.

- First, construct an overlattice tower $\mathcal{L}_0 \subset \cdots \subset \mathcal{L}_k$ that satisfies certain conditions.
- Then, enumerate all the points in $C_k = (\mathbf{t}_k + \mathcal{L}_k) \cap \mathcal{B}(\mathbf{0}, R_k)$.
- Subsequently, compute C_{i-1} iteratively, leveraging the points in C_i .
- Finally, the set of short vectors C_0 corresponding to the original lattice $\mathcal{L}$ can be obtained. Various lattice problems can then be easily solved using C_0 .

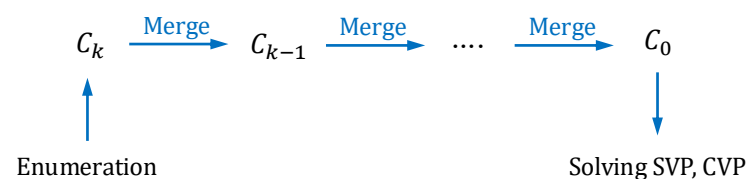


Figure 2. The overlattice-based method.

More specifically, the Overlattice-sieve algorithm encompasses an Overlattice-Tower subroutine, which is capable of returning bases $\mathbf{B}^{(i)}$ (for $i = 0, 1, \dots, k$) corresponding to a tower of overlattices $\mathcal{L} = \mathcal{L}_0 \subset \cdots \subset \mathcal{L}_k \subset \mathbb{R}^n$. This overlattice tower satisfies the following conditions:

- For each i , $\frac{\text{vol}(\mathcal{L}_i)}{\text{vol}(\mathcal{L}_{i-1})} = [\mathcal{L}_i : \mathcal{L}_{i-1}] = \lceil \alpha^n \rceil$.
- The lattice enumeration on $\mathcal{L}(\mathbf{B}^{(k)})$ is efficient.
- Given any two vectors $\mathbf{x}_1, \mathbf{x}_2 \in C_i$, it is efficient to determine whether $\mathbf{x}_1 + \mathbf{x}_2 \in C_{i-1}$.

Note that being “efficient” in the third item means being computable in polynomial time, whereas being “efficient” in the second item denotes being computable in single-exponential time. In particular, Becker, Gama, and Joux [29] proved that the bottom lattice

basis $\mathbf{B}^{(k)}$ derived from their proposed Overlattice-Tower subroutine is so-called “quasi-orthonormal”. Then, the time and space costs associated with the enumeration algorithm conducted on the quasi-orthonormal basis $\mathbf{B}^{(k)}$ do not exceed those of the subsequent Merge subroutine, in contrast to the super-exponential time complexity observed in general lattice enumeration scenarios.

On the other hand, the Overlattice-sieve algorithm also encompasses a Merge subroutine, which enables efficient computation of C_{i-1} using C_i . A naive method consists in constructing C_i by pairwise addition of vectors in $C_{i-1} \times C_{i-1}$. That is, for each pair of $\mathbf{x}_1, \mathbf{x}_2 \in C_{i-1}$, compute their sum $\mathbf{x}_1 + \mathbf{x}_2$ and check whether it belongs to C_{i-1} . However, this incurs a time complexity of $(\beta^n)^2$. To improve efficiency, Becker, Gama, and Joux [29] proposed a parallelized approach leveraging the properties of lattice cosets. In particular, by the definition of the overlattice tower, the lattice quotient group $\mathcal{L}_i / \mathcal{L}_{i-1}$ contains $[\mathcal{L}_i : \mathcal{L}_{i-1}] \approx \alpha^n$ cosets. A partition of C_i can be obtained by translating these cosets by $\mathbf{t}_i$. These translated cosets are henceforth referred to as “buckets”. Given that there are α^n buckets within C_i , the elements of C_i can be regrouped into α^n distinct buckets. Each element in C_i only requires pairing with elements from a single bucket, and each bucket contains approximately $\left(\frac{\beta}{\alpha}\right)^n$ elements. Consequently, each layer consumes a time complexity of approximately $\left(\frac{\beta^2}{\alpha}\right)^{n+o(n)}$. Overall, the time and space complexities of the Overlattice-sieve algorithm are $\left(\frac{\beta^2}{\alpha}\right)^{n+o(n)}$ and $\beta^{n+o(n)}$, respectively, provided that the following condition holds:

$$\beta \sqrt{1 - \left(\frac{\alpha}{2}\right)^2} \gtrsim 1, \quad (1)$$

Roughly speaking, the above inequality ensures that the Merge subroutine can retrieve all vectors in C_i at each layer, under the Gaussian Heuristic. See Figure 3 for an illustration. In this figure, the left and right circles are centered at $\mathbf{0}$ and $\mathbf{z}$, respectively, with a common radius of R_i . The shaded region of their intersection is denoted as I . In fact, for any point $\mathbf{z} \in C_{i-1}$, if a point $\mathbf{x} \in I$ can be found within the intersection region, then $\mathbf{y} = \mathbf{z} - \mathbf{x} \in C_i$. In other words, this simultaneously yields another point $\mathbf{y} \in C_i$ such that $\mathbf{x} + \mathbf{y} = \mathbf{z}$.

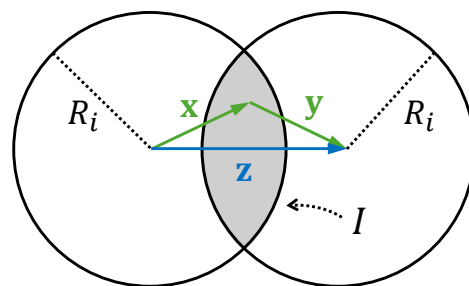


Figure 3. Merge subroutine. For any point $\mathbf{z} \in C_{i-1}$, if there exists a point $\mathbf{x} \in I$ within the intersection region, then there exists a point $\mathbf{y} = \mathbf{z} - \mathbf{x} \in C_i$.

To achieve this goal, we only need to guarantee that the intersection region $I = \mathcal{B}(\mathbf{0}, R_i) \cap \mathcal{B}(\mathbf{z}, R_i)$ contains at least one point $\mathbf{x} \in C_i$. Under the Gaussian Heuristic, it follows that we only need to ensure the volume of the intersection region is sufficiently large. That is,

$$\text{vol}(I) \geq G_H \cdot \text{vol}(\mathcal{L}_i).$$

From Corollary 2 in [29], we deduce that the volume of the intersection of two balls of radius R_i , with their centers separated by a distance of at most $R_{i-1} = \alpha R_i$, is greater than

$\frac{0.692}{\sqrt{n}} \cdot R_i^n \cdot \text{vol}\left(\mathcal{B}_n\left(\sqrt{1 - \left(\frac{\alpha}{2}\right)^2}\right)\right)$. A sufficient condition on the parameters α and β is then given by

$$R_i^n \cdot \frac{0.692}{\sqrt{n}} \cdot \text{vol}\left(\mathcal{B}_n\left(\sqrt{1 - \left(\frac{\alpha}{2}\right)^2}\right)\right) \geq G_H \cdot \text{vol}(\mathcal{L}_i).$$

Substituting the definition of R_i into the inequality yields

$$\left(\beta \cdot \sqrt{1 - \left(\frac{\alpha}{2}\right)^2}\right)^n \geq \frac{G_H \sqrt{n}}{0.692},$$

i.e.,

$$\beta \sqrt{1 - \left(\frac{\alpha}{2}\right)^2} \geq 1 + \varepsilon_n,$$

where $\varepsilon_n = \left(\frac{G_H \sqrt{n}}{0.692}\right)^{\frac{1}{n}} - 1$ is a small term that decreases to 0 as n tends to infinity. We thus arrive at Equation (1).

To optimize the time complexity of $\left(\frac{\beta^2}{\alpha}\right)^{n+o(n)}$, Becker, Gama, and Joux [29] set $\alpha = \sqrt{\frac{4}{3}}$ and $\beta = \sqrt{\frac{3}{2}}$, yielding the time and space complexities of the Overlattice-sieve algorithm as $2^{0.3774n}$ and $2^{0.2925n}$, respectively. For a detailed description and in-depth analysis of the aforementioned Overlattice-Tower and Merge subroutines, please refer to [29].

However, the limitation pertaining to the overlattice tower characterized by the index $[\mathcal{L}_i : \mathcal{L}_{i-1}] = \lceil \alpha^n \rceil$ and radius $R_i = \beta \cdot \text{gh}(\mathcal{L}_i)$ is rather artificial. It is not inherently necessary for the index of each layer within the overlattice tower to be fixed, nor is it justified for the radius of each layer to be a constant multiple of the Gaussian Heuristic. A natural inquiry thereby emerges:

If we treat the parameters α and β for each layer of the overlattice tower as flexible variables, denoted by α_i and β_i , respectively, can we improve the complexity of the Overlattice-sieve?

In the subsequent sections, we will present an almost negative response to this inquiry, thereby establishing a complexity lower bound for overlattice-based sieving algorithms.

4. Generalized Overlattice Tower Model

In Section 3, we observe that the Overlattice-sieve algorithm [29] directly fixes the index $[\mathcal{L}_i : \mathcal{L}_{i-1}]$ for all layers to a constant value when constructing the overlattice tower. On the other hand, the number of short vectors considered in each overlattice layer is also fixed. Technically, there is no inherent reason for these parameters to be constrained to constant values.

In this section, we analyze the overlattice-based method from a high-level perspective. We do not consider the details of enumeration on the bottom lattice, instead focusing on isolating the overlattice tower and analyzing the complexity of the Merge subroutine operating through this tower. We lift the restriction on fixed parameters for overlattice towers in overlattice-based methods and develop a generalized overlattice tower model.

In particular, for an overlattice tower $\mathcal{L} = \mathcal{L}_0 \subset \dots \subset \mathcal{L}_k \subset \mathbb{R}^n$, we set the following condition for each index i :

$$\frac{\text{vol}(\mathcal{L}_{i-1})}{\text{vol}(\mathcal{L}_i)} = (\alpha_{i-1})^n.$$

Under the Gaussian Heuristic, this condition directly yields the relation for each i :

$$\frac{\text{gh}(\mathcal{L}_{i-1})}{\text{gh}(\mathcal{L}_i)} = \alpha_{i-1}.$$

Additionally, we define

$$R_i = \beta_i \cdot \text{gh}(\mathcal{L}_i).$$

Then, under the Gaussian Heuristic, the size of the set C_i satisfies

$$|C_i| \approx \beta_i^n$$

for each i .

We adopt the analytical framework for the overlapping volume of two hyperballs as discussed in Section 3. Consider two n -dimensional hyperballs that are separated by a distance of $R_{i-1} = \beta_{i-1} \cdot \text{gh}(\mathcal{L}_{i-1})$, where both hyperballs possess a common radius of $R_i = \beta_i \cdot \text{gh}(\mathcal{L}_i)$. According to the Gaussian Heuristic, to ensure that the overlapping region of the two hyperballs contains at least one lattice point, the volume of their overlapping region, denoted as $\text{vol}(I)$, should be greater than the volume of the lattice $\mathcal{L}_i$, i.e.,

$$\text{vol}(I) \geq G_H \cdot \text{vol}(\mathcal{L}_i).$$

Upon scaling down both hyperballs by a factor of $\beta_i \cdot \text{gh}(\mathcal{L}_i)$, we arrive at two unit n -dimensional balls with a separation distance of $\frac{\beta_{i-1} \cdot \text{gh}(\mathcal{L}_{i-1})}{\beta_i \cdot \text{gh}(\mathcal{L}_i)} = \frac{\beta_{i-1}}{\beta_i} \cdot \alpha_{i-1}$. Denote the volume of the overlapping region of these two unit n -dimensional balls as $\text{vol}(I')$. From Corollary 2 in [29], we derive the following inequality:

$$\text{vol}(I') \geq \frac{0.692}{\sqrt{n}} \cdot \text{vol} \left(\mathcal{B}_n \left(\sqrt{1 - \left(\frac{\beta_{i-1}}{\beta_i} \cdot \frac{\alpha_{i-1}}{2} \right)^2} \right) \right).$$

Thus, it suffices to ensure that

$$\begin{aligned} \text{vol}(I) &\geq (\beta_i \cdot \text{gh}(\mathcal{L}_i))^n \cdot \frac{0.692}{\sqrt{n}} \cdot \text{vol} \left(\mathcal{B}_n \left(\sqrt{1 - \left(\frac{\beta_{i-1}}{\beta_i} \cdot \frac{\alpha_{i-1}}{2} \right)^2} \right) \right) \\ &\geq G_H \cdot \text{vol}(\mathcal{L}_i). \end{aligned}$$

That is,

$$\sqrt{\beta_i^2 - \frac{(\beta_{i-1})^2 (\alpha_{i-1})^2}{4}} \geq 1 + \varepsilon_n,$$

where $\varepsilon_n = \left(\frac{G_H \sqrt{n}}{0.692} \right)^{\frac{1}{n}} - 1$ is a small term that decreases towards 0 when the lattice dimension n grows.

To minimize the time complexity $\left(\frac{\beta_i^2}{\alpha_{i-1}} \right)^n$, we aim to maximize $\alpha_{i-1} \geq 1$ and minimize $\beta_i \geq 1$ simultaneously. Against this backdrop, to ensure the satisfaction of the aforementioned inequalities, we can explicitly set

$$\beta_i^2 - \frac{(\beta_{i-1})^2 (\alpha_{i-1})^2}{4} = 1^2,$$

i.e.,

$$\beta_i^2 = 1 + \frac{(\beta_{i-1})^2 (\alpha_{i-1})^2}{4}. \quad (2)$$

The core rationale lies in deriving the optimality-guaranteeing pair (α_0, β_1) starting from the initial condition $\beta_0 = 1$, followed by iteratively deriving the optimality-guaranteeing pair (α_{i-1}, β_i) from β_{i-1} until the final pair (α_{k-1}, β_k) is obtained. Herein, the term “overall optimality” pertains to the minimization of the total time complexity. Specifi-

cally, under the premise that $k = O(\text{poly}(n))$, our objective is to minimize the expression given by

$$\left(\frac{\beta_1^2}{\alpha_0}\right)^n + \left(\frac{\beta_2^2}{\alpha_1}\right)^n + \cdots + \left(\frac{\beta_k^2}{\alpha_{k-1}}\right)^n \approx O\left(\max_i \left(\frac{\beta_i^2}{\alpha_{i-1}}\right)^n\right).$$

That is, we aim to compute the minimum value of $\max_i \left(\frac{\beta_i^2}{\alpha_{i-1}}\right)^n$, where $\alpha_0, \alpha_1, \dots, \alpha_{k-1}$ and $\beta_0, \beta_1, \dots, \beta_k$ are variables subject to Equation (2). In the following section, we address how to solve this optimization problem. In particular, in Section 5, we will prove that its minimum value is $\frac{3\sqrt{3}}{4} \approx 2^{0.3774}$ with $\alpha_i = \sqrt{\frac{4}{3}}$ and $\beta_i = \sqrt{\frac{3}{2}}$ for all i , consistent with the original Overlattice-sieve [29]. Note that the optimal parameter sets $\{\alpha_i, \beta_i\}_i$ are not unique, but all yield the same asymptotic complexity.

5. Lower Bound on Overlattice-Sieve

In this section, we conduct an in-depth analysis of the time complexity lower bound for the overlattice-based method, based on the generalized overlattice tower model proposed in Section 4. Our analysis reveals that the original Overlattice-sieve [29] has already attained the complexity lower bound under this framework. Accordingly, none of the sieving algorithms under this framework can surpass the BDGL-sieve [27] that adopts the LSF technique.

In the following, we conduct a comprehensive parameter analysis of the generalized overlattice tower model. In order to simplify the exposition, we define

$$\psi(\alpha_{i-1}) = \frac{\beta_i^2}{\alpha_{i-1}}.$$

Note that in the definition of the function ψ , we designate α_{i-1} as the independent variable. This is because, for fixed values of α_{i-1} and β_{i-1} , we can derive β_i by virtue of Equation (2). We then combine this β_i with the newly assigned α_i to obtain the subsequent β_{i+1} from the same equation. Through this recursive procedure, we yield a sequence of optimal values for $\alpha_0, \alpha_1, \dots, \alpha_{k-1}$ and $\beta_0, \beta_1, \dots, \beta_k$.

Recall that our primary objective is to minimize $\max_{i \in \{1, \dots, k\}} \psi(\alpha_{i-1})$. First, for a fixed index i , we investigate the minimum value of $\psi(\alpha_{i-1})$. By substituting Equation (2) into the definition of $\psi(\alpha_{i-1})$, the following is obtained:

$$\begin{aligned} \psi(\alpha_{i-1}) &= \frac{\beta_i^2}{\alpha_{i-1}} \\ &= \frac{1 + \frac{(\beta_{i-1})^2(\alpha_{i-1})^2}{4}}{\alpha_{i-1}} \\ &= \frac{1}{\alpha_{i-1}} + \frac{(\beta_{i-1})^2}{4} \alpha_{i-1}. \end{aligned}$$

Treating β_{i-1} as a constant, the above expression of $\psi(\alpha_{i-1})$ exhibits the characteristic form of a hook function. This function achieves its minimum at $\alpha_{i-1} = \frac{2}{\beta_{i-1}^2}$, with the corresponding minimum value being

$$\psi\left(\frac{2}{\beta_{i-1}^2}\right) = \beta_{i-1}.$$

Nevertheless, directly setting $\alpha_{i-1} = \frac{2}{\beta_{i-1}}$ leads to

$$\beta_i^2 = 1 + \frac{(\beta_{i-1})^2(\alpha_{i-1})^2}{4} = 2,$$

which proves to be a highly undesirable situation. This is because in the next iteration, the value of β_{i+1}^2 satisfies the inequality

$$\begin{aligned}\beta_{i+1}^2 &= 1 + \frac{(\beta_i)^2(\alpha_i)^2}{4} \\ &> 1 + \frac{2^2 \cdot 1^2}{4} = 2,\end{aligned}$$

which is excessively large. Furthermore, for all subsequent iterations $j \geq i$, the inequality $\beta_j \geq 2$ holds true, thereby resulting in

$$\alpha_{i-1} = \frac{2}{\beta_{i-1}} \leq 1.$$

This ultimately renders the construction of the overlattice tower unfeasible.

Therefore, it is impossible for $\psi(\alpha_{i-1})$ to attain its minimum value for all i . We choose $\alpha_{i-1} < \frac{2}{\beta_{i-1}}$ rather than $\alpha_{i-1} > \frac{2}{\beta_{i-1}}$, as our objective is to minimize $\beta_i^2 = 1 + \frac{(\beta_{i-1})^2(\alpha_{i-1})^2}{4}$ in the subsequent iteration as much as possible.

5.1. Fixed α and Flexible β_i

To simplify the analysis, we first investigate a special class of cases: when

$$1 < \alpha_0 = \alpha_1 = \dots = \alpha_{k-1} < 2.$$

Here, the condition $1 < \alpha_i$ arises from the essential requirement of the overlattice tower, while the condition $\alpha_i < 2$ is imposed by technical constraints, as it ensures that the series $\{\beta_i\}_i$ forms a convergent geometric sequence.

We first consider the case when $i = 0$, for which we have

$$\begin{aligned}\psi(\alpha_0) &= \frac{\beta_1^2}{\alpha_0} \\ &= \frac{1 + \frac{(\beta_0)^2(\alpha_0)^2}{4}}{\alpha_0} \\ &= \frac{1}{\alpha_0} + \frac{1}{4}\alpha_0.\end{aligned}$$

Subsequently, we get

$$\begin{aligned}\beta_1^2 &= \psi(\alpha_0) \cdot \alpha_0 \\ &= 1 + \frac{(\alpha_0)^2}{4}.\end{aligned}$$

We then examine the case for $i = 1$, yielding

$$\begin{aligned}\psi(\alpha_1) &= \frac{\beta_2^2}{\alpha_1} \\ &= \frac{1 + \frac{(\beta_1)^2(\alpha_1)^2}{4}}{\alpha_1} \\ &= \frac{1}{\alpha_0} + \frac{1}{4}\alpha_0 + \frac{1}{16}(\alpha_0)^3,\end{aligned}$$

where the last equality holds due to our assumption that all α_i are identical. We then have

$$\begin{aligned}\beta_2^2 &= \psi(\alpha_1) \cdot \alpha_1 \\ &= 1 + \frac{1}{4}(\alpha_0)^2 + \frac{1}{16}(\alpha_0)^4.\end{aligned}$$

We claim the following statement holds.

Theorem 1. For any $i \in \{0, 1, \dots, k-1\}$, we have

$$\begin{aligned}\psi(\alpha_i) &= \frac{1}{\alpha_0} + \frac{1}{4}(\alpha_0)^1 + \frac{1}{4^2}(\alpha_0)^3 + \dots + \frac{1}{4^{i+1}}(\alpha_0)^{2(i+1)-1}, \\ \beta_{i+1}^2 &= 1 + \frac{1}{4}(\alpha_0)^2 + \frac{1}{4^2}(\alpha_0)^4 + \dots + \frac{1}{4^{i+1}}(\alpha_0)^{2(i+1)}.\end{aligned}$$

Proof. Mathematical induction can be readily applied to prove this statement. We have already established the validity of the statement for the cases where $i = 0$ and $i = 1$. Suppose we have already established that, for all $j < i$,

$$\begin{aligned}\beta_j^2 &= 1 + \frac{1}{4}(\alpha_0)^2 + \frac{1}{4^2}(\alpha_0)^4 + \dots + \frac{1}{4^j}(\alpha_0)^{2j} \\ \psi(\alpha_{j-1}) &= \frac{1}{\alpha_0} + \frac{1}{4}(\alpha_0)^1 + \frac{1}{4^2}(\alpha_0)^3 + \dots + \frac{1}{4^j}(\alpha_0)^{2j-1}.\end{aligned}$$

As a matter of fact, we have

$$\begin{aligned}\psi(\alpha_i) &= \frac{\beta_{i+1}^2}{\alpha_i} \\ &= \frac{1 + \frac{(\beta_i^2)(\alpha_i)^2}{4}}{\alpha_i} \\ &= \frac{1 + \frac{\left[1 + \frac{1}{4}(\alpha_0)^2 + \frac{1}{4^2}(\alpha_0)^4 + \dots + \frac{1}{4^i}(\alpha_0)^{2i}\right] \cdot (\alpha_i)^2}{4}}{\alpha_i} \\ &= \frac{1}{\alpha_i} + \frac{1}{4} \left[1 + \frac{1}{4}(\alpha_0)^2 + \frac{1}{4^2}(\alpha_0)^4 + \dots + \frac{1}{4^i}(\alpha_0)^{2i}\right] \alpha_i \\ &= \frac{1}{\alpha_0} + \frac{1}{4}(\alpha_0)^1 + \frac{1}{4^2}(\alpha_0)^3 + \dots + \frac{1}{4^{i+1}}(\alpha_0)^{2(i+1)-1}.\end{aligned}$$

Therefore, it holds that

$$\beta_{i+1}^2 = \psi(\alpha_i) \cdot \alpha_i = 1 + \frac{1}{4}(\alpha_0)^2 + \frac{1}{4^2}(\alpha_0)^4 + \dots + \frac{1}{4^{i+1}}(\alpha_0)^{2(i+1)}.$$

This concludes the proof. $\square$

Consequently, the overall complexity can be controlled solely by the single parameter α_0 . In fact, for a fixed α_0 , both $\psi(\alpha_{k-1})$ and β_k^2 form monotonically increasing bounded sequences with respect to k and thus must converge to their limits. Their respective limits are given by

$$\begin{aligned}\beta_k^2 &= 1 + \left(\frac{\alpha_0^2}{4}\right)^1 + \left(\frac{\alpha_0^2}{4}\right)^2 + \dots + \left(\frac{\alpha_0^2}{4}\right)^k \rightarrow \frac{4}{4 - \alpha_0^2} \quad (k \rightarrow \infty); \\ \psi(\alpha_{k-1}) &= \frac{\beta_k^2}{\alpha_{k-1}} \rightarrow \frac{4}{4\alpha_0 - \alpha_0^3} \quad (k \rightarrow \infty).\end{aligned}$$

Consequently, the term $\frac{4}{4\alpha_0 - \alpha_0^3}$ for $\alpha_0 \in (1, 2)$ attains its minimum value of $\frac{3\sqrt{3}}{4} \approx 2^{0.37744}$ at $\alpha_0 = \sqrt{\frac{4}{3}}$. In this case, we have $\lim_{k \rightarrow \infty} \beta_k^2 = \frac{3}{2} \approx 2^{2 \cdot 0.2925}$. It is noteworthy that the time complexity $2^{0.37744n}$ and space complexity $2^{0.2925n}$ yielded by this set of parameters are identical to those of the original Overlattice-sieve.

5.2. General Case: Flexible α_i, β_i

We next extend our analysis to the general case. We first consider the case when $i = 0$, for which we have

$$\begin{aligned}\psi(\alpha_0) &= \frac{\beta_1^2}{\alpha_0} \\ &= \frac{1 + \frac{(\beta_0)^2(\alpha_0)^2}{4}}{\alpha_0} \\ &= \frac{1}{\alpha_0} + \frac{1}{4}\alpha_0.\end{aligned}$$

Subsequently, we obtain

$$\begin{aligned}\beta_1^2 &= \psi(\alpha_0) \cdot \alpha_0 \\ &= 1 + \frac{(\alpha_0)^2}{4}.\end{aligned}$$

We then examine the case for $i = 1$, which yields

$$\begin{aligned}\psi(\alpha_1) &= \frac{\beta_2^2}{\alpha_1} \\ &= \frac{1 + \frac{(\beta_1)^2(\alpha_1)^2}{4}}{\alpha_1} \\ &= \frac{1}{\alpha_1} + \frac{1}{4} \left[1 + \frac{(\alpha_0)^2}{4} \right] \alpha_1 \\ &= \frac{1}{\alpha_1} + \frac{1}{4}\alpha_1 + \frac{1}{4^2}(\alpha_0)^2\alpha_1.\end{aligned}$$

It follows that

$$\begin{aligned}\beta_2^2 &= \psi(\alpha_1) \cdot \alpha_1 \\ &= 1 + \frac{1}{4}(\alpha_1)^2 + \frac{1}{4^2}(\alpha_1)^2(\alpha_0)^2.\end{aligned}$$

Similar to Theorem 1, we can employ mathematical induction to prove the following statement, whose proof is omitted here for brevity.

Theorem 2. For any $i \in \{0, 1, \dots, k-1\}$, we have

$$\begin{aligned}\psi(\alpha_i) &= \frac{1}{\alpha_i} + \frac{1}{4}\alpha_i + \frac{1}{4^2}\alpha_i(\alpha_{i-1})^2 + \dots + \frac{1}{4^{i+1}}\alpha_i(\alpha_{i-1})^2 \dots (\alpha_0)^2, \\ \beta_{i+1}^2 &= 1 + \frac{1}{4}(\alpha_i)^2 + \frac{1}{4^2}(\alpha_i)^2(\alpha_{i-1})^2 + \dots + \frac{1}{4^{i+1}}(\alpha_i)^2(\alpha_{i-1})^2 \dots (\alpha_0)^2.\end{aligned}$$

We do not directly compute the minimum value of $\psi(\alpha_i)$ in the general case. Instead, we establish the following key lemma. By virtue of this lemma, we develop an approach to derive a tight lower bound on the computational complexity of the overlattice-based method.

Lemma 1. For all i , we only need to consider the case where $\beta_{i+1} \geq \beta_i$.

Proof. We proceed by contradiction, assuming that there exists some index i such that $\beta_{i+1} < \beta_i$. Recall that the baseline time complexity of the Overlattice-sieve algorithm is $\frac{3\sqrt{3}}{4}$. Thus, we only need to consider scenarios where the time complexity is less than or equal to $\frac{3\sqrt{3}}{4}$. Formally, this means

$$\frac{\beta_{i+1}^2}{\alpha_i} \leq \frac{3\sqrt{3}}{4},$$

or equivalently,

$$\alpha_i \geq \frac{4}{3\sqrt{3}}\beta_{i+1}^2.$$

This implies the following chain of inequalities:

$$\begin{aligned}\beta_{i+1}^2 &= 1 + \frac{(\alpha_i\beta_i)^2}{4} \\ &\geq 1 + \frac{\beta_i^2}{4} \left(\frac{4}{3\sqrt{3}}\beta_{i+1}^2 \right)^2 \\ &> 1 + \frac{\beta_{i+1}^2}{4} \left(\frac{4}{3\sqrt{3}}\beta_{i+1}^2 \right)^2.\end{aligned}$$

We now analyze the resulting univariate high-degree inequality:

$$4\beta_{i+1}^6 - 27\beta_{i+1}^2 + 27 < 0.$$

Let $x := \beta_{i+1}^2$. The inequality then reduces to

$$(x+3)(-2x+3)^2 < 0,$$

which is impossible since $x = \beta_{i+1}^2 > 0$. This yields a contradiction. Hence, there is no index i for which $\beta_{i+1} < \beta_i$. This concludes the proof. $\square$

Therefore, we can assume without loss of generality that $\beta_{i+1} \geq \beta_i$ for all $i \in \{0, 1, \dots, k-1\}$ by Lemma 1. On the other hand, to ensure that the time complexity of the overlattice-based sieve algorithm is less than or equal to that of the original variant, the series $\{\beta_i\}_i$ must be bounded. Formally speaking, we have $\beta_i \leq \frac{3\sqrt{3}}{4}$ for all i . Otherwise, the overall time complexity of the algorithm would exceed that of the original BGJ-sieve, as at least β_i^n lattice points in each level of the overlattice tower must be enumerated. By Lemma 1, we know that the sequence $\{\beta_i\}_i$ is monotonically increasing and bounded, and thus it must converge. We may without loss of generality assume that it converges to β_∞ . In other words, for sufficiently large i , $\beta_i \approx \beta_\infty$. Thus, for sufficiently large i , we may set $\beta_i = \beta_{i+1}$, which gives

$$\beta_i^2 = \beta_{i+1}^2 = 1 + \frac{(\alpha_i\beta_i)^2}{4}.$$

Rearranging this expression yields

$$\beta_i^2 = \frac{1}{1 - \frac{1}{4}\alpha_i^2} = \beta_{i+1}^2.$$

Substituting the value of β_{i+1}^2 into the above, we obtain

$$\begin{aligned}\psi(\alpha_i) &= \frac{\beta_{i+1}^2}{\alpha_i} \\ &= \frac{1}{\alpha_i \left(1 - \frac{1}{4}\alpha_i^2\right)} \\ &= \frac{4}{-\alpha_i(\alpha_i + 2)(\alpha_i - 2)}.\end{aligned}$$

To minimize $\psi(\alpha_i)$, it suffices to maximize its denominator, or equivalently, to solve $\operatorname{argmin}_{\alpha_i \in (1,2)} f(\alpha_i)$ where

$$f(\alpha_i) = -\alpha_i(\alpha_i + 2)(\alpha_i - 2).$$

The function $\psi(\alpha_i)$ attains its minimum value of $\frac{3\sqrt{3}}{4} \approx 2^{0.3774}$ when we set $\alpha_i = \sqrt{\frac{4}{3}}$ and $\beta_i = \sqrt{\frac{3}{2}}$.

Once again, this retrieves the complexity result corresponding to the original Overlattice-sieve. In summary, even when we treat the parameters α and β in the original overlattice tower as flexible variables α_i and β_i , we still cannot yield any improvement in the time complexity of this method. This proves that the original Overlattice-sieve algorithm already attains the complexity lower bound for overlattice-based sieving methods.

6. Conclusions

This work focuses on the time complexity analysis of overlattice-based sieving methods, a branch of lattice sieving algorithms for solving the SVP and CVP. A key contribution of this work is the establishment of a tight lower bound for the time complexity of overlattice-based sieving. Specifically, we prove that the time complexity of the original Overlattice-sieve algorithm [29] reaches this lower bound of $2^{0.3774n}$, indicating that the original algorithm is theoretically optimal within the existing overlattice-based sieving framework.

This lower bound result further clarifies the performance gap between overlattice-based methods and state-of-the-art lattice sieving algorithms. Compared with the BDGL-sieve [27], which achieves $2^{0.292n+o(n)}$ time complexity, our findings confirm that overlattice-based sieving methods cannot surpass the time complexity of the BDGL-sieve under the current technical framework. When integrated with the lower bound result by Kirshanova and Laarhoven [30], which demonstrates that the BDGL-sieve has attained optimal performance within the NNS-based sieving paradigm, our conclusions collectively define the performance ceiling of existing lattice sieving algorithms. Specifically, no incremental optimization within the current technical frameworks can achieve a fundamental breakthrough in time complexity. Any substantial improvement will require revolutionary innovations in the core paradigms or key techniques of lattice sieving.

Finally, our findings strengthen the theoretical foundation for the current concrete security evaluation of lattice-based cryptosystems. They validate the rationality of the current standard practice that adopts the BDGL-sieve's complexity as the benchmark for assessing the core SVP method. Given that both overlattice-based and NNS-based sieving paradigms have reached their respective performance limits, reliance on the BDGL-sieve's complexity to guide cryptographic parameter selection and security margin evaluation is well founded, providing valuable support for the development and standardization of post-quantum cryptography.

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Abbreviations

The following abbreviations are used in this manuscript:

PQC	post-quantum cryptography
NNS	Nearest Neighbor Search
SVP	Shortest Vector Problem
CVP	Closest Vector Problem
LSF	Locality-Sensitive Filters
LSH	Locality-Sensitive Hashing

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